

End Semester Examination 2025

Name of the course: B. Tech (ECE)

Semester: VI

Name of the Paper: VLSI Technology and Design

Paper Code: **TEC 603**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two **questions** among a, b, & c *in each main question*.
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	What is Integrated Circuit? Explain the advantages of ICs.	CO1
(b)	In which form is Silicon available in the Earth? Explain the techniques for the extraction of single crystal Si from the polycrystalline Si.	
(c)	Explain in brief about thin oxide and thick oxide, their process of deposition and applications in VLSI.	
Q2	(20 marks)	
(a)	Explain in brief about lift off process in VLSI.	CO2
(b)	Explain Physical vapor deposition techniques for metallization applications in VLSI.	
(c)	What do you understand by Negative Photoresists and Positive Photoresists and their applications in VLSI.	
Q3	(20 marks)	
(a)	Consider an n – channel metal oxide semiconductor field effect transistor (MOSFET) with gate to source voltage of 1.5V. Assume that $W/L = 4$, $\mu_n C_{ox} = 50 \times 10^{-6} \text{ A/V}^2$, and the threshold voltage is 0.5V. In the saturation region, find the drain conduction g_d (in micro-Siemens).	CO3
(b)	Derive the expression for current vs voltage for MOS in non-saturation and saturation mode.	
(c)	The slope of the I_D vs. V_{GS} curve of an n – channel MOSFET in linear region is $10^{-4} \Omega^{-1}$ at $V_{DS} = 0.2V$. For the same device, neglecting channel length modulation, the slope of the $\sqrt{I_D}$ vs V_{GS} curve (in $\sqrt{AV}$) under saturation region is approximately.	
Q4	(20 marks)	
(a)	Explain in detail about Inverters with n-type MOSFET load.	CO4
(b)	Explain briefly about switching power Dissipation in CMOS inverter.	
(c)	Explain Switching characteristics of MOS inverters, and Delay-time definitions.	
Q5	(20 marks)	
(a)	Draw the stick diagram of expression $\{(A+B)E+CD\}'$	CO5/6
(b)	Draw step by step layout for complement of $\{(A+B)C\}'$.	
(c)	What are the various industrial applications of CMOS? Explain briefly.	

